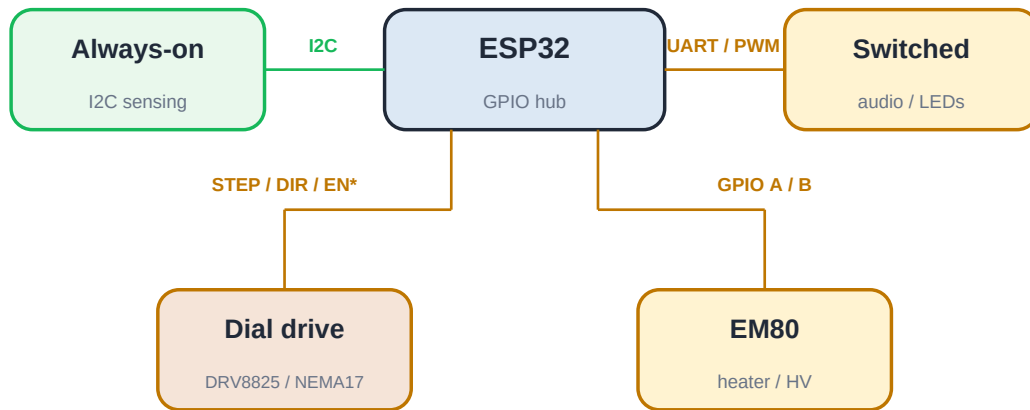


1. System wiring at a glance

Follow named connectors; power rails and signal roles use a consistent colour key.

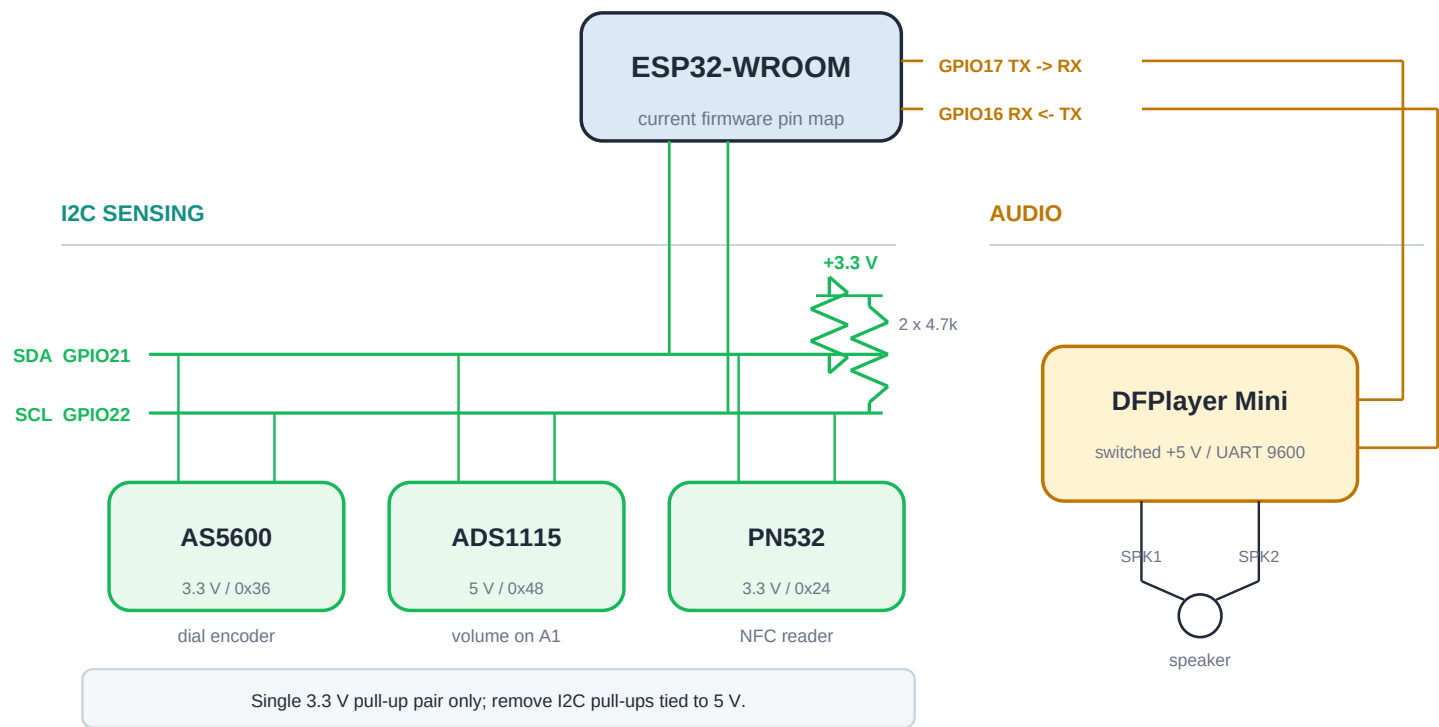


* DRV8825 pins are firmware placeholders - confirm physical wiring.

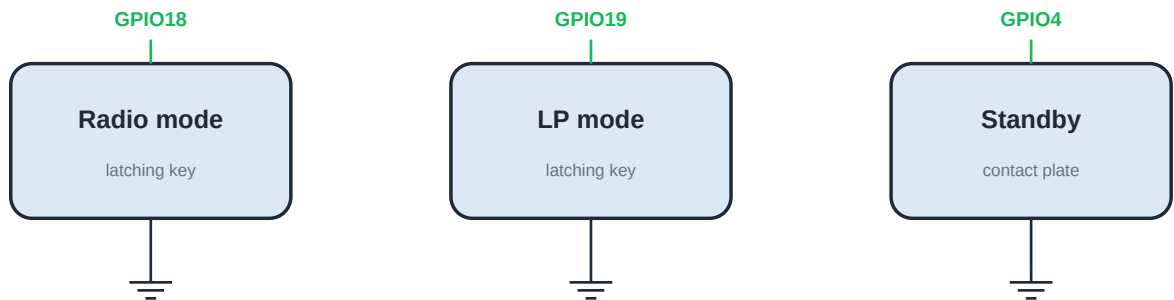
Role	Colour	Meaning
Power	Red +5 V / amber HV	Supply rails; connect only to the labelled rail
ESP / signal	Green / charcoal	GPIO, I2C, UART and control connections
Ground	Charcoal	All low-voltage modules share common GND

2. Controls, sensing and audio

Current low-voltage wiring, grouped by function with no shared signal crossings.



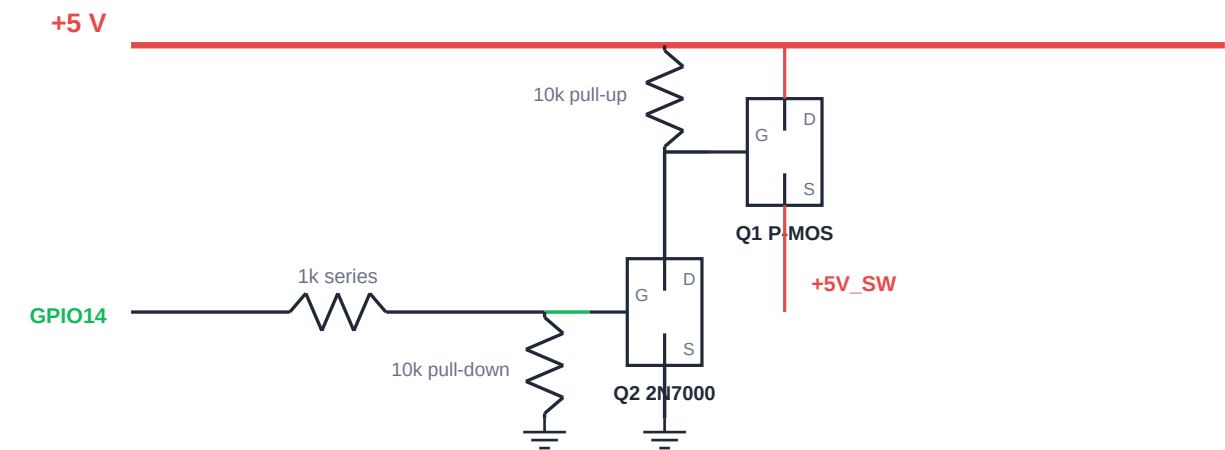
FRONT-PANEL INPUTS



Module / input	Power	Interface	Operational note
AS5600	always-on 3.3 V	I2C 0x36	tracked in standby
ADS1115	always-on 5 V	I2C 0x48 / A1	I2C stays 3.3 V-safe
PN532	3.3 V	I2C 0x24	reader in I2C mode
DFPlayer Mini	switched +5 V	UART 9600	TX/RX crossed
Radio / LP / standby	ESP32 pull-ups	GPIO18 / 19 / 4	contacts close to GND

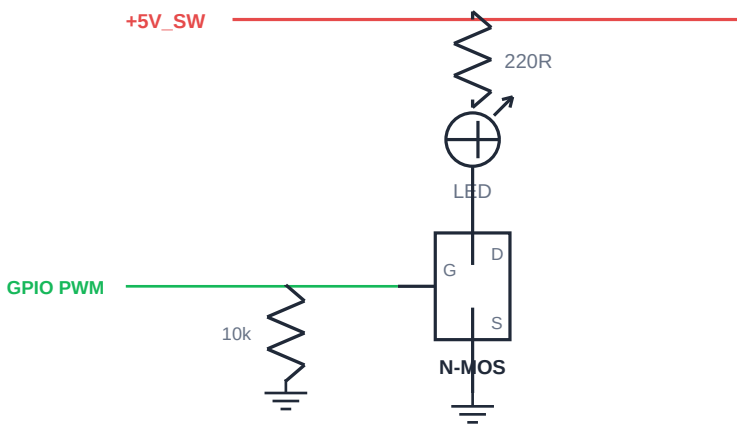
3. Power rail, LEDs and standby

A compact connected schematic for the switched rail and one repeated LED driver stage.



GPIO14 HIGH turns Q2 on, pulls Q1 gate LOW and enables +5V_SW; GPIO LOW leaves the rail OFF.

LED low-side driver - repeat this stage x5

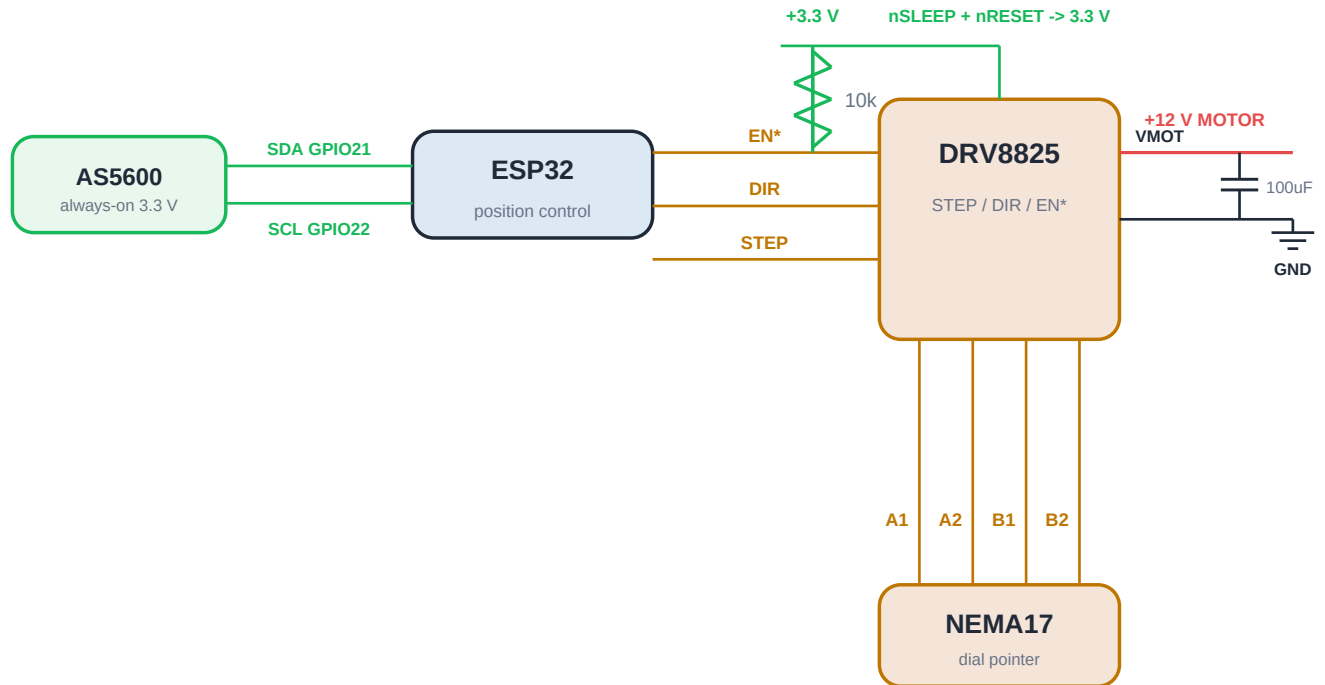


LED1 GPIO27 / LED2 GPIO25 / LED3 GPIO32 / LED4 GPIO33 / LED5 GPIO26
Each channel gets its own 220R resistor, N-MOS stage and 10k gate pull-down.

Circuit	ESP32 control	Required note
Switched rail	GPIO14	HIGH enables via Q2 level shifter; never PWM
LED channels 1-5	GPIO27 / 25 / 32 / 33 / 26	One MOSFET stage per LED
Standby plate	GPIO4	Plate to GND = active; open = standby

4. Dial motor and encoder

Keep motor power separate from ESP32 logic; confirm the driver's ground arrangement.



DRV8825 pin	Firmware mapping	Build requirement
STEP	GPIO13*	Provisional - verify before flashing
DIR	GPIO5*	Provisional - verify before flashing
EN (active low)	GPIO15*	Add 10k pull-up to 3.3 V for coils-off boot
VMOT / GND	external 12 V	100 uF electrolytic directly at the driver
Logic inputs	3.3 V signals	No VDD pin; ESP32 and driver share GND
nSLEEP / nRESET	tie high together	Required for STEP pulses to operate

Safety

Set the DRV8825 current limit conservatively. Never connect or disconnect the motor while powered.